MANNALA ABIRAM

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PROFESSIONAL SUMMARY

AI/ML Engineer and Computer Science graduate (2026) with hands-on experience building Generative AI, Large Language Model (LLM), and Retrieval-Augmented Generation (RAG) systems using Python, LangChain, and NLP. Built an AI Resume Shortlister that automated candidate evaluation and reduced manual screening effort by 60%, alongside machine learning and data analysis projects using Pandas, NumPy, and Scikit-learn. Strong foundation in Python, SQL, Git, and software engineering practices, with a track record of independently designing, building, and deploying end-to-end AI-driven applications. Seeking an AI/ML Engineer or Machine Learning Intern role to apply strong problem-solving, analytical thinking, and collaboration in a fast-paced AI engineering team.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript

AI / ML / Generative AI: Machine Learning, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn ,LangChain ,LangGraph

Databases: PostgreSQL, SQLAlchemy

Developer Tools: Git, GitHub, Jupyter Notebook

Data Analysis & Visualization: Exploratory Data Analysis (EDA), Data Cleaning, Data Visualization, Statistical Analysis

Other: JWT Authentication, SDLC, MS Excel

PROJECTS

AI Resume Shortlister – LLM + RAG-Based Screening System GitHub

- Designed and built an end-to-end resume screening system using **Python**, **LangChain**, **NLP**, and **RAG** to automate candidate evaluation against job descriptions.
- Engineered a PDF-parsing and semantic resume-to-job matching pipeline capable of processing **500+ resumes**.
- Automated candidate ranking and shortlisting logic, reducing manual screening effort by **60%**.

MediQ – Full-Stack Healthcare Appointment & Management Platform Live

- Built a full-stack healthcare platform enabling doctor search, online appointment booking, and patient management.
- Designed a scalable **FastAPI** backend with **PostgreSQL** and **SQLAlchemy** to manage hospitals, doctors, appointments, and patient records.
- Implemented **JWT**-based authentication and role-based modules for admins, doctors, and patients.
- Followed **SDLC** best practices to deliver a modular, maintainable architecture across frontend and backend.

Interactive Data Analysis & Visualization Web Application GitHub

- Developed a **Flask**-based web application for interactive data exploration and visualization.
- Automated data cleaning, **EDA**, and dashboard generation using **Pandas** and **Matplotlib** across datasets of **10,000+ records**.
- Reduced manual data analysis time by **50%** through reusable, automated analysis pipelines.

EDUCATION

B.Tech in Computer Science Engineering Expected Apr 2026

Anantha Lakshmi Institute of Technology & Sciences, JNTU Anantapur

CGPA: 7.14/10

CERTIFICATIONS

- Data Analysis & Decision Making (NPTEL)
- Artificial Intelligence Fundamentals (IBM)

ADDITIONAL INFORMATION

- Strong analytical, problem-solving, and software engineering foundation, with hands-on experience across AI-driven and full-stack applications.
- Adaptable, collaborative team player with a continuous learning mindset and active interest in Generative AI, LLMs, and Retrieval-Augmented Generation .